

IPHONE R/C CAR

SENIOR PROJECT



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1. INTRODUCTION

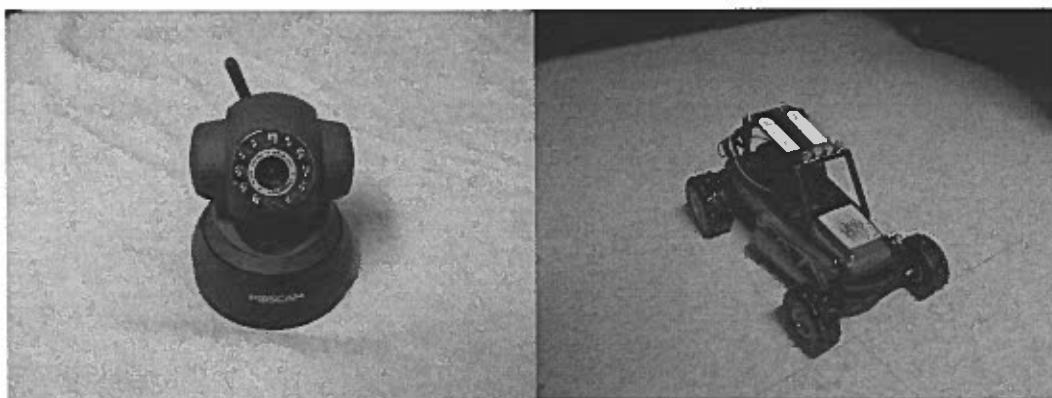
- a. This project will use an iPhone/iPod Touch to view live video feed from a wireless video camera and control the movement from the Radio Controlled (R/C) Car.
- b. This project is useful for situations in case of emergency that professionals may need. Paramedics can use a RC Car in critical moments to save lives and reduce injuries and deaths.

2. PROJECT BACKGROUND AND PURPOSE

- a. The purpose of this project is to use an iPhone/iPod Touch to control a remote control car and, in general, to learn how to create applications for the iPhone.
- b. The reason we chose to do this was because we had a lot of interest in developing an application for the iPhone. Also, since the iPhone is really popular, we wanted to learn how to develop for it.

3. PROJECT SUMMARY

- a. We gathered the required materials:
 - i. Web Cam
 - ii. RC Car
 - iii. iPod Touch
 - iv. Router





- b. The webcam is a FOSCAM FI8908 by Shenzhen Foscam Intelligent Technology Co., Ltd. It is a wireless network camera that allows you to connect to the camera wirelessly or through Ethernet. It features a pan/tilt feature and allows for remote monitoring when connected to a network.
- c. The RC Car is a remote controlled product that uses RF technology for communication.
- d. The router is a D-link Wireless router that allows wireless-N networking (Wi-Fi a/b/g also) and Ethernet connection. Up to 4 Ethernet devices can connect to the router.
- e. The device we ran our program on comes from the Apple iPod Touch. It is the 4th generation iPod product that includes 8GB storage and Wi-Fi connectivity (wireless a/b/g/n). Similar to the iPhone and iPad, the iPod touch runs on the same iOS ecosystem that allows our application to be compatible on all three devices.
- f. We created a program using Xcode SDK that features the user interface that we are going to use. The user interface uses the accelerometer to control the car. It also has buttons that the user can interact with.
- g. We also are able to get the camera to successfully stream video to the iPhone/iPod Touch.
- h. We were planning on installing a circuit board, but we were not able to get the circuit board because of lack of funds.

4. RESEARCH

a. Objective-C Programming

- i. Our programming language of choice is Objective-C. The Objective-C language is a simple computer language designed to enable sophisticated object-oriented programming. This language is recommended for iPhone programming but there are other programming languages that can be used to develop iPhone applications.

b. Apple Xcode for iPhone programming

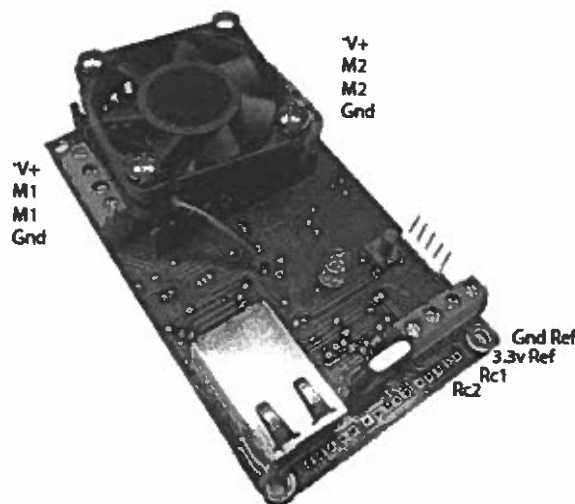
- i. The program we used to develop our iPhone application was Apple Xcode. Xcode, along with iOS SDK, is the complete toolset for building MAC OS X and iOS applications.

c. Video Streaming

- i. Tracking the location of the R/C car can be used by the wireless camera. We wanted to add a neat feature when tracking the R/C Car without the naked eye. The camera we're using is the FOSSCAM Wireless Camera. It has the ability stream video wirelessly or wired (with Ethernet).

d. Circuit Board (Barracuda Wi-Fi robot controller) - Communication between iPhone and R/C car

- We intended to use a circuit board to communicate between the R/C car and the iPhone. Without the circuit board, the R/C car can only communicate through its remote controller. The Barracuda Wi-Fi robot controller is a piece of hardware that connects to the R/C car to receive signals from devices. Compatible devices may include a computer, smartphone, or any other product that is program to the hardware. The circuit board has an Ethernet port that connects to a router for wireless communication. Signals received from the iPhone can be sent through the router to the circuit board where it can perform movement specified.



Due to lack of funds, we were not able to implement this hardware to our project but this is one of the items needed to control an R/C car from the iPhone.

5. DESIGN

b. UI Design

- Included with this project is an iPhone application that adds usability to the R/C car. Inside the application, the UI includes direction indicators, an go button to accelerate the car, and a window screen to view video feed from the camera. The direction indicators are used to show current direction of R/C car movement. If the iPhone is in a straight position, then both arrows will be greyed out. Else if the iPhone is tilted left or right, the arrows will light to the appropriate direction. This feature uses the iPhone accelerometer that detects the current angle position the device is in.
- The video streaming window is used to show video feed from the camera. The user can follow the R/C car by viewing the video on the iPhone, even when the car is not in sight. This feature eliminates the need to follow the car to determine its location. The user can simply see where the car is going from the video feed.



c. Video Streaming from a Web Browser

- The small window that is used for video streaming is made from a web browser window. To access the camera video feed, you must have the device IP address, username, and password. All three of these points are placed in the web address to gain proper access. Using the web browser is the only choice to access the camera.

7. PROBLEMS ENCOUNTERED

a. Video Streaming operation

- The problem with video streaming dealt with HTTP authentication to login. Instead of using the regular IP web address for access we used a special web address to bypass the security login window. But there are limitations when using the special IP address. Determining which circuit board to use
- When using the special IP address, the iPhone was not able to use the camera's advanced features including
 - i. Pan/Tilt feature
 - ii. Access to camera settings

- We also had an issue with video size in the program. We were not able to enlarge the video size to fit web browser window. Here are some a few things we tested.
 - i. Change the video size in Interface Builder
 - ii. Use `CGRectMake()` to alter video size

b. Circuit Board

- Our hope was to manage the R/C car through the iPhone but due to a lack of funds, we were not able to do so.

APPENDICES

● PROJECT CODE

```
//
//  PrototypeAppDelegate.h
//  Prototype
//
//  Created by Justin Jenkins on 12/15/10.
//  Copyright 2010 __MyCompanyName__. All rights reserved.
//

#import <UIKit/UIKit.h>

@class PrototypeViewController;

@interface PrototypeAppDelegate : NSObject <UIApplicationDelegate> {
    UIWindow *window;
    PrototypeViewController *viewController;
}

@property (nonatomic, retain) IBOutlet UIWindow *window;
@property (nonatomic, retain) IBOutlet PrototypeViewController
*viewController;

@end
```

```

//
//  PrototypeAppDelegate.m
//  Prototype
//
//  Created by Justin Jenkins on 12/15/10.
//  Copyright 2010 __MyCompanyName__. All rights reserved.
//

#import "PrototypeAppDelegate.h"
#import "PrototypeViewController.h"

@implementation PrototypeAppDelegate

@synthesize window;
@synthesize viewController;

#pragma mark -
#pragma mark Application lifecycle

- (BOOL)application:(UIApplication *)application
didFinishLaunchingWithOptions:(NSDictionary *)launchOptions {

    // Override point for customization after application launch.

    // Add the view controller's view to the window and display.
    [window addSubview:viewController.view];
    [window makeKeyAndVisible];

    return YES;
}

- (void)applicationWillResignActive:(UIApplication *)application {
    /*
     Sent when the application is about to move from active to
     inactive state. This can occur for certain types of temporary
     interruptions (such as an incoming phone call or SMS message) or when
     the user quits the application and it begins the transition to the
     background state.
     Use this method to pause ongoing tasks, disable timers, and
     throttle down OpenGL ES frame rates. Games should use this method to
     pause the game.
     */
}

```

```
- (void)applicationDidEnterBackground:(UIApplication *)application {
    /*
       Use this method to release shared resources, save user data,
       invalidate timers, and store enough application state information to
       restore your application to its current state in case it is terminated
       later.
       If your application supports background execution, called instead
       of applicationWillTerminate: when the user quits.
    */
}
```

```
- (void)applicationWillEnterForeground:(UIApplication *)application {
    /*
       Called as part of transition from the background to the inactive
       state: here you can undo many of the changes made on entering the
       background.
    */
}
```

```
- (void)applicationDidBecomeActive:(UIApplication *)application {
    /*
       Restart any tasks that were paused (or not yet started) while the
       application was inactive. If the application was previously in the
       background, optionally refresh the user interface.
    */
}
```

```
- (void)applicationWillTerminate:(UIApplication *)application {
    /*
       Called when the application is about to terminate.
       See also applicationWillDidEnterBackground:.
    */
}
```

```
#pragma mark -
#pragma mark Memory management
```

```
- (void)applicationDidReceiveMemoryWarning:(UIApplication
*)application {
```

```
    /*  
    Free up as much memory as possible by purging cached data objects  
    that can be recreated (or reloaded from disk) later.  
    */  
}
```

```
- (void)dealloc {  
    [viewController release];  
    [window release];  
    [super dealloc];  
}
```

```
@end
```

```
//  
//  PrototypeViewController.h  
//  Prototype  
//  
//  Created by Justin Jenkins on 12/15/10.  
//  Copyright 2010 __MyCompanyName__. All rights reserved.  
//  
  
#import <UIKit/UIKit.h>  
  
@interface PrototypeViewController : UIViewController {  
    IBOutlet UIButton *goButton;  
  
}  
  
-(IBAction) pushedGo:(id) sender;  
  
@end
```

```

//
//  PrototypeViewController.m
//  Prototype
//
//  Created by Justin Jenkins on 12/15/10.
//  Copyright 2010 __MyCompanyName__. All rights reserved.
//

#import "PrototypeViewController.h"
#import "WebViewController.h"

@implementation PrototypeViewController

- (BOOL)shouldAutorotateToInterfaceOrientation:
(UIInterfaceOrientation)interfaceOrientation {
    // Return YES for all supported orientations
    return (interfaceOrientation ==
UIInterfaceOrientationLandscapeLeft);
}

- (void)didReceiveMemoryWarning {
    // Releases the view if it doesn't have a superview.
    [super didReceiveMemoryWarning];

    // Release any cached data, images, etc that aren't in use.
}

- (void)viewDidUnload {
    // Release any retained subviews of the main view.
    // e.g. self.myOutlet = nil;
}

-(IBAction) pushedGo:(id) sender {
    //NSURL *url = [NSURL
URLWithString:@"http://192.168.0.10/videostream.cgi?user=admin2&pwd=pw
d&"];
    NSURL *url = [NSURL URLWithString:@"http://192.168.0.10/"];
    WebViewController *webViewController = [[WebViewController alloc]
initWithURL:url];
    [self presentModalViewController:webViewController animated:YES];
    //[[webViewController loadRequest:url]];
}

```

```
        [webViewController release];  
    }  
  
    - (void)dealloc {  
        [goButton release];  
        [super dealloc];  
    }  
  
@end
```

```

//
// WebViewController.h
// Prototype
//
// Created by Justin Jenkins on 12/15/10.
// Copyright 2010 __MyCompanyName__. All rights reserved.
//

#import <UIKit/UIKit.h>

@interface WebViewController : UIViewController <UIWebViewDelegate,
UIAccelerometerDelegate> {

    NSURL *theURL;
    NSString *theTitle;
    IBOutlet UIWebView *webView;
    IBOutlet UINavigationController *webTitle;

    IBOutlet UIImageView *gasPedal;
    IBOutlet UIImageView *redArrow;

    IBOutlet UILabel *yCoordinate;
    IBOutlet UILabel *yDirection;
    NSTimer *touchTimer;

    IBOutlet UILabel *xLabel;
    IBOutlet UILabel *yLabel;
    IBOutlet UILabel *zLabel;

    IBOutlet UILabel *yStatusLabel;

    IBOutlet UIProgressView *xBar;
    IBOutlet UIProgressView *yBar;
    IBOutlet UIProgressView *zBar;

    UIAccelerometer *accelerometer;

    IBOutlet UIImageView *leftArrow;
    IBOutlet UIImageView *rightArrow;

}

- (IBAction) done:(id)sender;
- (IBAction) gasPedalPressed:(id)sender;

```

```
- (IBAction) gasPedalReleased:(id)sender;
@property (nonatomic, retain) UIWebView *webView;
@end
```

```
//
//  WebViewController.m
//  Prototype
//
//  Created by Justin Jenkins on 12/15/10.
//  Copyright 2010 __MyCompanyName__. All rights reserved.
//
```

```
#import "WebViewController.h"
```

```
@implementation WebViewController
```

```
#pragma mark UIAccelerometer Methods
```

```
//accelerometer function for direction of RC Car
```

```
- (void)accelerometer:(UIAccelerometer *)meter
```

```
didAccelerate:(UIAcceleration *)acceleration {
```

```
    yCoordinate.text = [NSString stringWithFormat:@"%f",
acceleration.y];
```

```
    if (acceleration.y < -0.2) {
        yDirection.text = @"LEFT";
        [leftArrow setImage: [UIImage
imageName:@"leftArrow.png"]];
        [rightArrow setImage:[UIImage
imageName:@"inactive_rightArrow.png"]];
    }
    else if (acceleration.y > 0.2) {
        yDirection.text = @"RIGHT";
        [rightArrow setImage: [UIImage
imageName:@"rightArrow.png"]];
        [leftArrow setImage: [UIImage
imageName:@"inactive_leftArrow.png"]];
    }
    else {
        yDirection.text = @"NEUTRAL";
        [rightArrow setImage: [UIImage
imageName:@"inactive_rightArrow.png"]];
        [leftArrow setImage: [UIImage
imageName:@"inactive_leftArrow.png"]];
    }
}
```

```

//Window title - removed
- (id)initWithURL:(NSURL *)url andTitle:(NSString *)string {
    if( self = [super init] ) {
        theURL = url;
        theTitle = string;
    }
    return self;
}

- (id)initWithURL:(NSURL *)url {
    return [self initWithURL:url andTitle:nil];
}

// Implement viewDidLoad to do additional setup after loading the
view, typically from a nib.
- (void)viewDidLoad {
    [super viewDidLoad];
    //webTitle.title = theTitle;
    NSURLRequest *requestObject = [NSURLRequest
requestWithURL:theURL];
    [webView loadRequest:requestObject];

    accelerometer = [UIAccelerometer sharedAccelerometer];
    accelerometer.updateInterval = 0.1;
    accelerometer.delegate = self;
    //accelerometer = [UIAccelerometer sharedAccelerometer];
    //accelerometer.updateInterval = 0.1;
    //accelerometer.delegate = self;
}

- (IBAction) done:(id)sender {
    [self dismissModalViewControllerAnimated:YES];
}

- (IBAction) gasPedalPressed:(id)sender {
    //redArrow = FALSE;
    [redArrow setImage: [UIImage imageNamed:@"red_arrow.png"]];
}

```

```

- (IBAction) gasPedalReleased:(id)sender {
    //redArrow = FALSE;
    [redArrow setImage: [UIImage
imageNamed:@"inactive_upArrow.png"]];
}

- (BOOL)shouldAutorotateToInterfaceOrientation:
(UIInterfaceOrientation)interfaceOrientation {
    // Return YES for all supported orientations
    //return YES;
    return (interfaceOrientation ==
UIInterfaceOrientationLandscapeLeft);
}

- (void)didReceiveMemoryWarning {
    // Releases the view if it doesn't have a superview.
    [super didReceiveMemoryWarning];

    // Release any cached data, images, etc that aren't in use.
}

- (void)viewDidUnload {
    [super viewDidUnload];
    // Release any retained subviews of the main view.
    // e.g. self.myOutlet = nil;
}

- (void)dealloc {
    [webView release];
    [webTitle release];
    [super dealloc];

    [leftArrow release];
    [rightArrow release];
    [yDirection release];
    [yCoordinate release];
}

@end

```

Rescue RC Car

Justin Jenkins
Josiah Vergonio

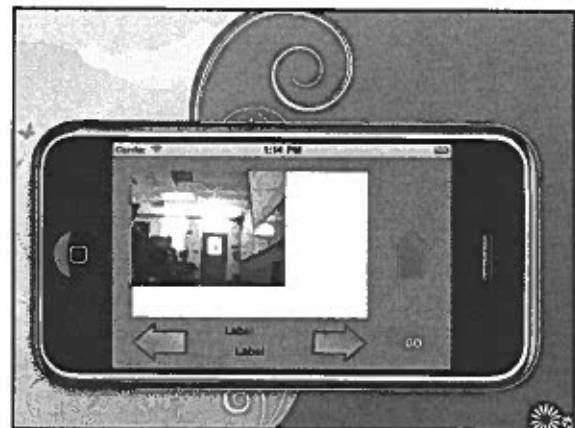
Advisor: Dr. Lin

Introduction

- This project will use an iPhone/iPod Touch to view live video feed from a wireless video camera and control the movement from the Radio Controlled (R/C) Car.
- Why? This project is useful for situations in case of emergency that professionals may need. Paramedics can use a RC Car in critical moments to save lives and reduce injuries and deaths.

Project Background / Purpose

- a. The purpose of this project is to use an iPhone/iPod Touch to control a remote control car and, in general, to learn how to create applications for the iPhone.
- b. The reason we chose to do this was because we had a lot of interest in developing an application for the iPhone. Also, since the iPhone is really popular, we wanted to learn how to develop for it.



Problems encountered in the project

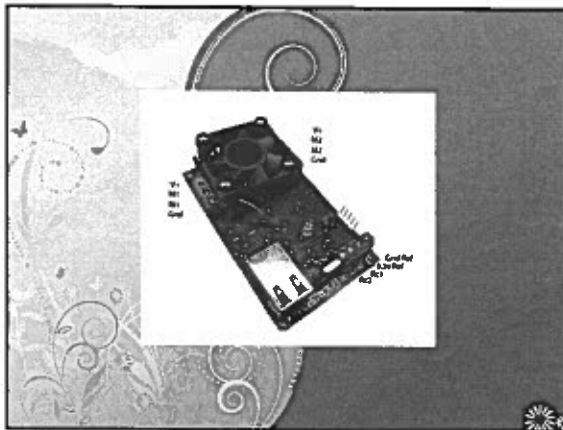
- UI Direction Arrows
- Video Feed
- Circuit Board

UI Direction Arrows

UI direction arrows were a challenge but yet simple to fix once we found a solution.

In Xcode, the solution was to implement the UI elements programmatically than adding it directly to Interface Builder program.





Theory

- Use the accelerometer on the iPhone to control the RC car.
 - Write a program on the XCode SDK that utilizes the accelerometer class
 - Create the user interface with the Interface Builder. The Interface Builder is a tool within the XCode that helps developers create user interfaces.
- Install the circuit board inside the car, the router is placed on the car. The camera is plugged into the router.
- The program will communicate with the circuit board.
- The circuit board is connected to a wireless router via an Ethernet port.
- The wireless router wirelessly connects to the iPhone for communication.

Theory

- The program will communicate with the circuit board and camera via the wireless router.
- Then, the circuit board control the car.

Conclusion

- Our application was able to use the accelerometer and camera.
- Our hope was to manage the R/C car through the iPhone.